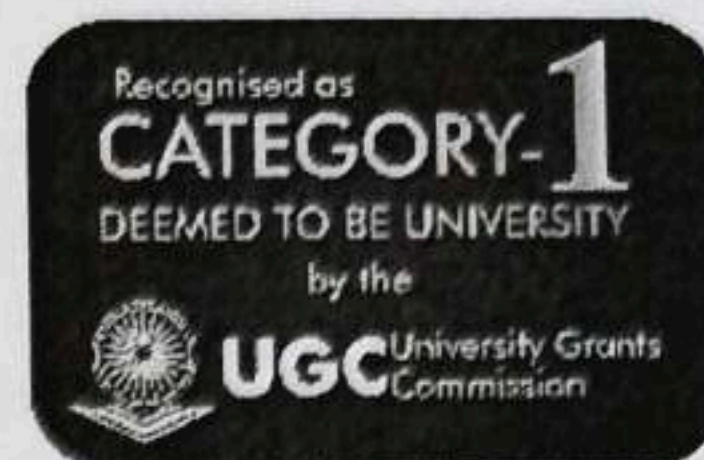




Vel Tech

Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)



School of Computing
Department of CSE (Data Science)
ACADEMIC YEAR 2025 – 2026 (Summer Semester)

BONAFIDE CERTIFICATE

NAME: P. Koteswara Rao

VTU NO: VTU28129

REG.NO: 24UEDLOO53

BRANCH: B.Tech CSE (DATA SCIENCE)

YEAR/SEM: 2nd year / 3rd sem

SLOT: S15-L12

Certified that this is a bonafide record of work done by above student in the
"10211DS207 – Database Management Systems" during the year 2025-2026.

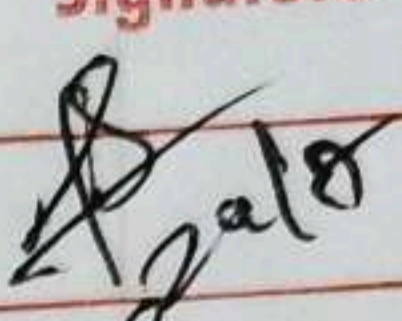
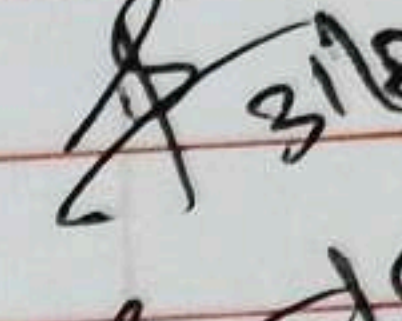
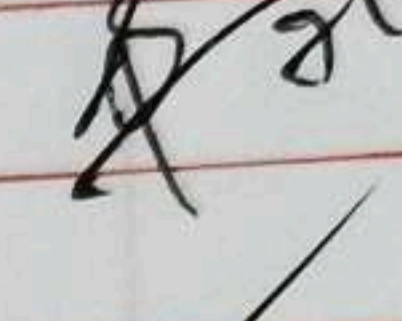
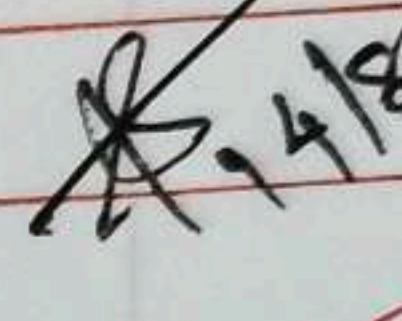
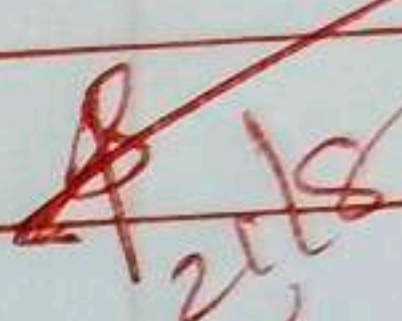
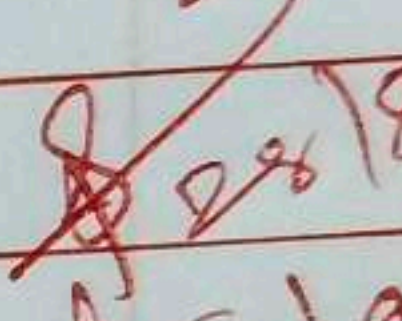
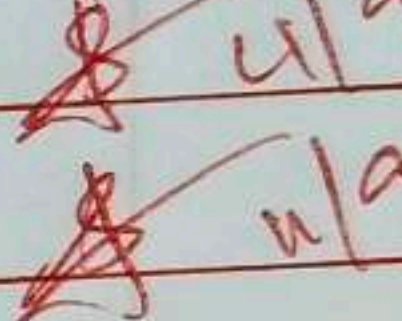
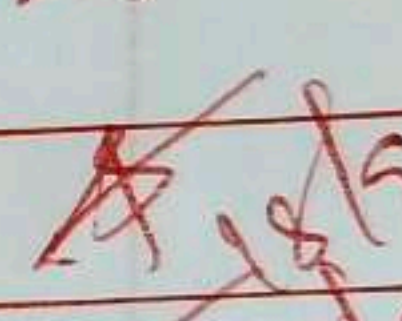
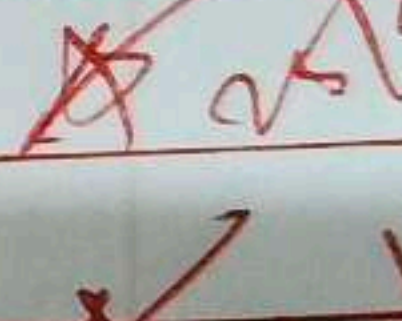
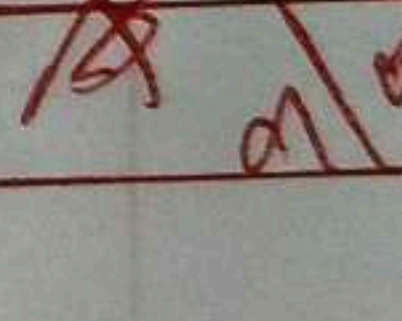
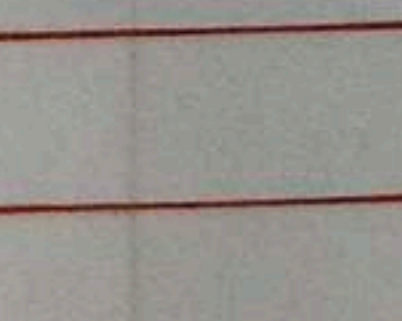
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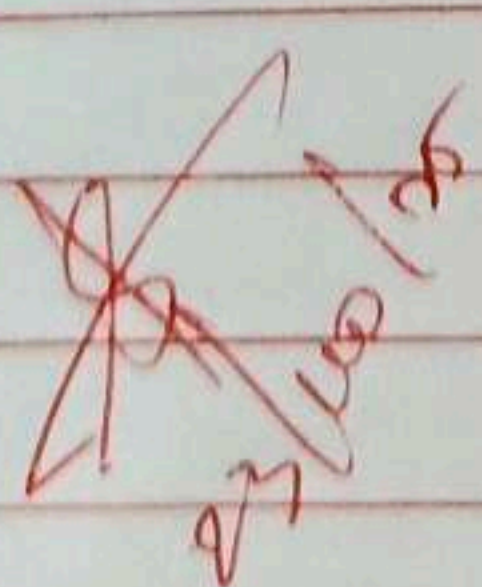
SIGNATURE OF INCHARGE

Submitted for the Semester Model Examination held on _____ at Vel Tech
Rangarajan Dr. Sagunthala R & D Institute of Science and Technology.

EXAMINER 1

EXAMINER 2

S.No.	Date	Topic	Page No. MARKS	Signature
1.	24/07/2025	conceptual design AFTER ETR	20	
2.	24/07/2025 31/7/2025	Generating design of other traditional database models	20	
3.	07/08/2025	Developing Queries with DML single-row Functions and operations	19	
4.	04/08/2025	Developing Queries with DML Multi-row Functions and operations	20	
5.	09/08/2025	writing Join Queries, Equivalency, and Recursive Queries	20	
6.	20/08/2025	PL/SQL Procedures, Functions, Loops	19	
7.	4/09/2025	PL/SQL Procedure for loops	20	
8.	11/09/2025	Normalizing databases using functional dependencies up to BCNF	20	
9.	18/09/2025	Backing up and recovery in databases	20	
10.	25/09/2025	CRUD operations in document databases	20	
11.	01/10/2025	CRUD operations in Graph databases	20	

Completed


TASK-1

31/07/2025

A Bank Management system is a comprehensive software solution designed to manage and streamline banking operations. It covers various aspects of banking, including customer account management, transaction processing, loan and mortgage management and more.

AIM:- To draw Conceptual Design through FTR using drawio tool.

Procedure:-

Step 1(a): Identifying the Entities

Entities represent the major objects in the banking system.

Sample output:-

- * customer
- * Account
- * Transaction
- * Loan
- * Branch
- * Employee

Step 1(b): Identifying the Attributes

For each entity, list out the attributes

(primary key is underlined)

Sample output:-

1. customer

→ customer ID (PK)

→ Name

→ Address

→ Phone

→ email

2. Account

→ Account - Number (PK)

→ Account - Type

→ Balance

→ Date - opened

3. Transaction

→ Transaction - ID (PK)

→ Transaction - Type

→ Amount

→ Date - Time

4. Loan

→ Loan - ID (PK)

→ Loan - Type

→ Loan - Amount

→ Interest - Rate

5. Branch

→ Branch - ID (PK)

→ Branch - Name

→ Location

6. Employee

→ Employee - ID (PK)

→ Name

→ Position

→ Contact - No

Step 1 (c): Identifying Relationships, Cardinality and Type

1. Customer - Account

* A customer has one or more accounts

* Cardinality: One to Many.

* Type: strong relationship

2. Account - Transaction

* An account performs many transactions

* cardinality: One to Many (1:M)

* Type: strong relationship

3. customer_loan

- * A customer takes zero or many loans
- * cardinality : One to Many (1:M)
- * Type : strong relationship

4. Branch_employee

- * A Branch employs many employees
- * Cardinality : One to Many.

5. Branch_Account

- * A Branch manages many accounts
- * Cardinality : One to Many (1:M)

Sample output:-

- customer (1) -- (M) Account
- Account (1) -- (M) Transaction
- customer (1) -- (M) Loan.
- Branch (1) -- (M) Employee
- Branch (1) -- (M) Account

Step 1(a): Reframing the relations with key and constraints.

Convert the conceptual model into relations (tables) with keys

Sample output

1. Customer (customer_ID [PK], Name, Address, phone, Email)
2. Branch (Branch_ID [PK], Branch_Name, location)
3. Account (Account_Number [PK], Account_Type, Balance, Date-opened, customer_ID [FK], Branch_ID [FK])
4. Transaction (Transaction_ID [PK], Transaction_Type, Amount, Date-time, Account_Number [FK])
5. Loan (Loan_ID [PK], Loan_Type, Loan_Amount, Interest-Rate, Start-date, customer_ID [FK])

6. Employee (Employee_ID [PK], Name, Position, Contact_No, Branch_ID [FK])

Constraints:

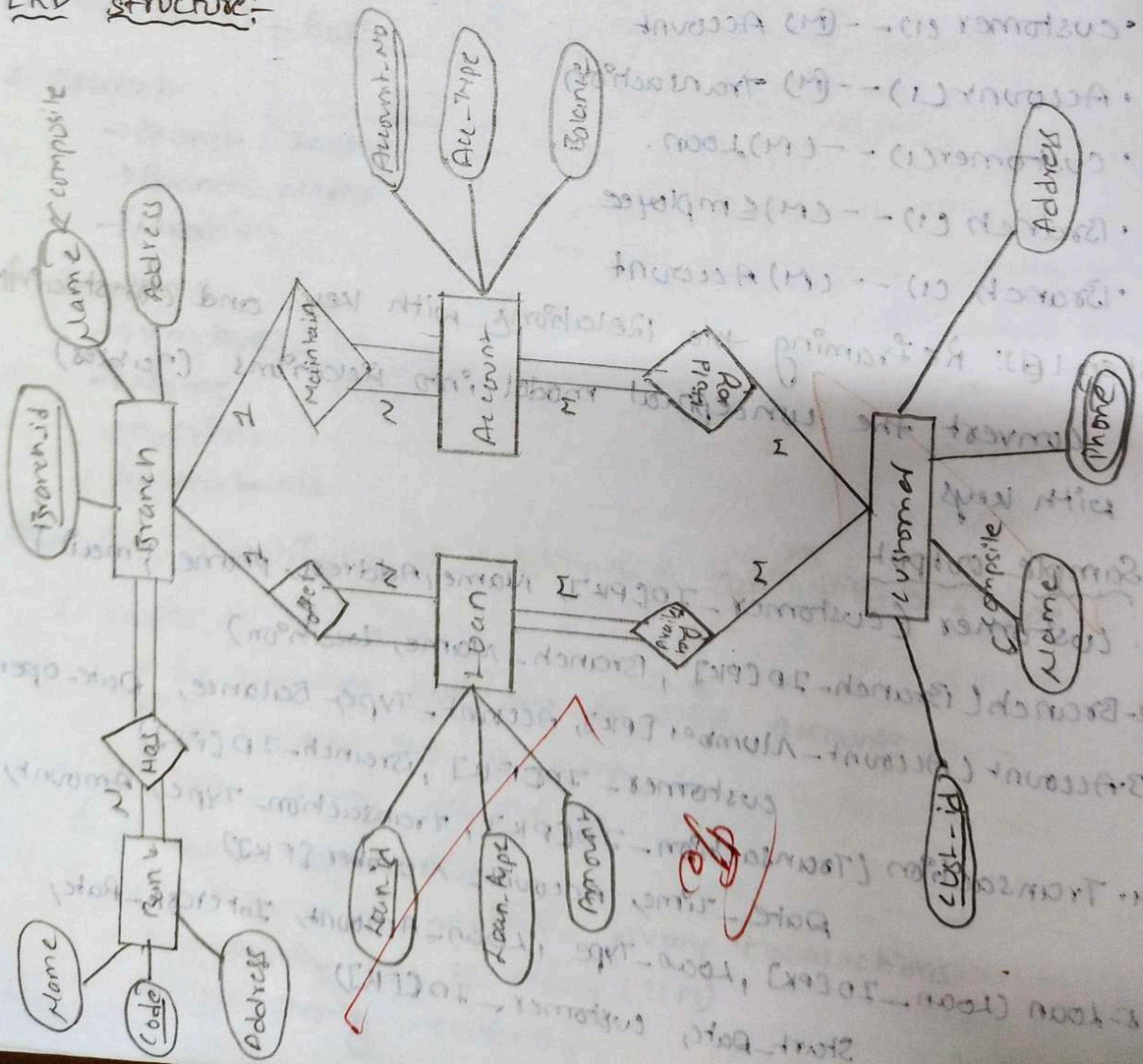
- PK: Unique identifiers for each table.
- FK: Foreign keys maintain referential integrity
- TM: Enforced using foreign key relationships.

step(1c): Develop IER Diagram

By using draw.io to represent:

- Entities as rectangles
- Attributes as ovals
- Relationships as diamonds
- Indicate cardinalities (1:M)

ERD structure:-



THU 22

Comparing design of other hierarchical database model
existing hierarchical database model of the database by
creating the data model of the database by
creating the data model of the database by

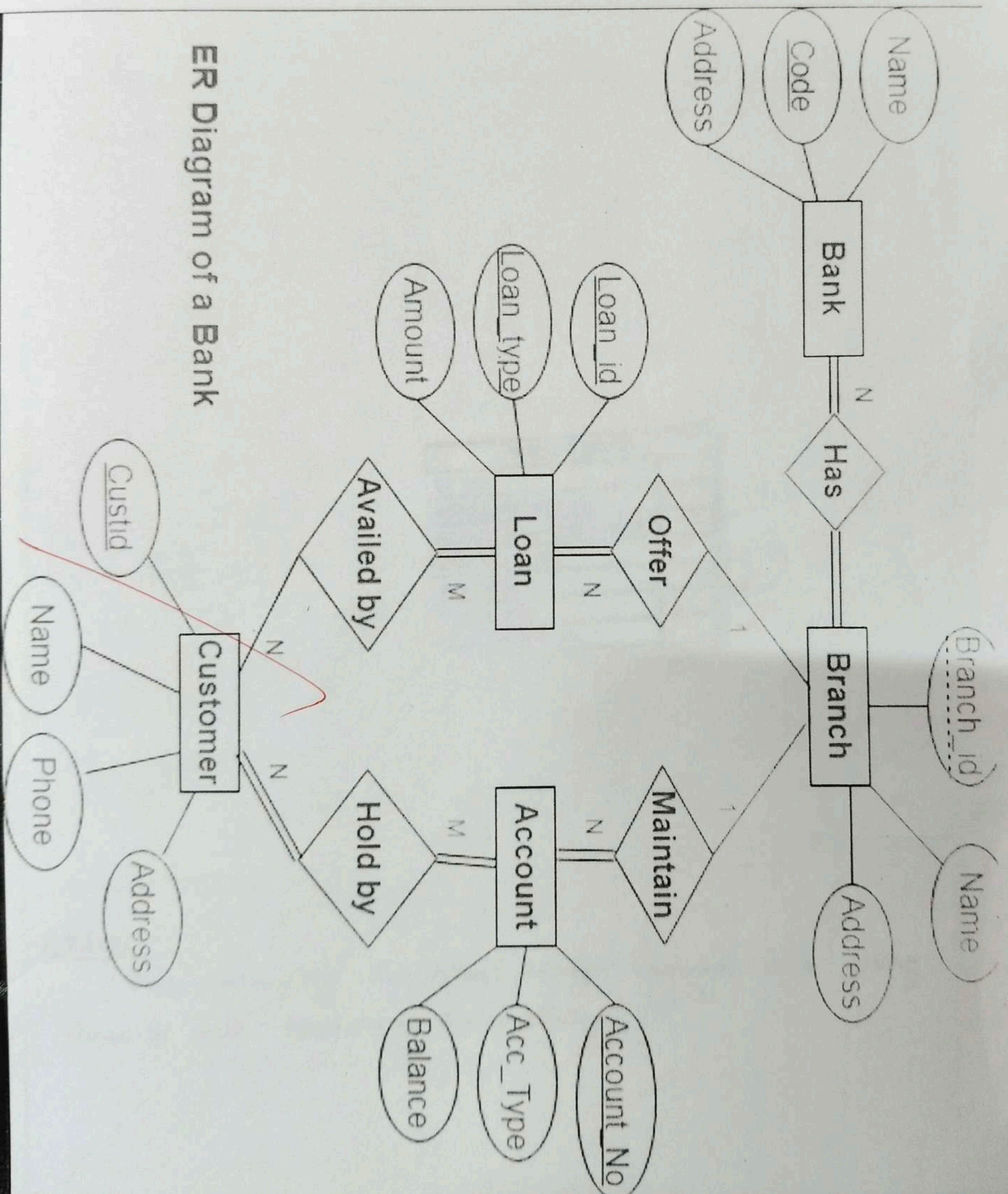
Identifying the relationship of each relationship, line and
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EX NO.	1
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	20
DATE WITH DATE	20/11/20

RESULT:-

Thus, drawing conceptual design through FTR using
draw.io was executed successfully



ER Diagram of a Bank